

Homework Assignment 3: Criticality in the S_N Approximation

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The method

All three questions solve the same monoenergetic, steady-state problem,

$$\mu \frac{\partial \psi}{\partial x} + \Sigma_t \psi = \frac{1}{2} \left[\Sigma_s \phi + \frac{1}{k} \nu \Sigma_f \phi \right], \quad \phi(x) = \int_{-1}^1 \psi(x, \mu) d\mu \quad (1)$$

with the same three ingredients: a Gauss–Legendre quadrature in angle, diamond differencing in space with a negative-flux fixup, and the Bell & Glasstone k iteration. Lengths are mean free paths in Questions 3 and 4 and centimetres in Question 5.

Angle. The N ordinates $\{\mu_m\}$ and weights $\{w_m\}$ are the Gauss–Legendre nodes on $[-1, 1]$, so that $\mu_{N-m+1} = -\mu_m$, $\sum_m w_m = 2$, and

$$\phi_i = \sum_{m=1}^N w_m \psi_{i,m} \quad (2)$$

Space, slab. Integrating over cell i and closing with the diamond relation $\psi_{i,m} = \frac{1}{2}(\psi_{i+1/2,m} + \psi_{i-1/2,m})$ gives the sweep

$$\psi_{i,m} = \frac{2|\mu_m| \psi_{\text{in},m} + \frac{1}{2} S_i \Delta x}{\Sigma_t \Delta x + 2|\mu_m|}, \quad \psi_{\text{out},m} = 2\psi_{i,m} - \psi_{\text{in},m} \quad (3)$$

swept from $x = a/2$ inwards for $\mu_m < 0$ and from $x = 0$ outwards for $\mu_m > 0$.

Space, sphere. The spherical equation carries an angular redistribution term,

$$\mu \frac{\partial \psi}{\partial r} + \frac{1 - \mu^2}{r} \frac{\partial \psi}{\partial \mu} + \Sigma_t \psi = \frac{S}{2} \quad (4)$$

whose conservative form integrates over the cell to

$$\mu_m \left(A_{i+\frac{1}{2}} \psi_{i+\frac{1}{2},m} - A_{i-\frac{1}{2}} \psi_{i-\frac{1}{2},m} \right) + \frac{A_{i+\frac{1}{2}} - A_{i-\frac{1}{2}}}{w_m} \left(\alpha_{m+\frac{1}{2}} \psi_{i,m+\frac{1}{2}} - \alpha_{m-\frac{1}{2}} \psi_{i,m-\frac{1}{2}} \right) + \Sigma_t V_i \psi_{i,m} = \frac{S_i}{2} V_i \quad (5)$$

with $A = 4\pi r^2$ and $V_i = \frac{4\pi}{3}(r_{i+1/2}^3 - r_{i-1/2}^3)$. This is (3) with a second outgoing face, in angle, closed by the same diamond relation $\psi_{i,m} = \frac{1}{2}(\psi_{i,m+1/2} + \psi_{i,m-1/2})$. The angular coefficients follow from demanding that a flat flux solve (5) exactly:

$$\alpha_{\frac{1}{2}} = 0, \quad \alpha_{m+\frac{1}{2}} = \alpha_{m-\frac{1}{2}} - w_m \mu_m, \quad \alpha_{N+\frac{1}{2}} = 0 \quad \text{since} \quad \sum_m w_m \mu_m = 0 \quad (6)$$

The recursion needs a starting value $\psi_{i,1/2}$ at $\mu_{1/2} = -1$, where $1 - \mu^2 = 0$ and the angular derivative drops out; what remains, $-\partial\psi/\partial r + \Sigma_t\psi = S/2$, is the plain slab sweep at $|\mu| = 1$, run inwards from the vacuum boundary.

Fixup. Whenever a diamond outgoing flux $2\psi_{i,m} - \psi_{in,m}$ comes out negative it is set to zero and the cell balance is re-solved with it held there, in angle as well as in space. This never occurs in Questions 3–5: the fission source is proportional to the flux itself, so no cell is ever starved, and a critical system is thin enough that $\Sigma_t\Delta x/|\mu_m| < 2$ at every ordinate.

Boundary conditions. Reflective at the symmetry point, $\psi_{1/2,m} = \psi_{1/2,m'}$ with $\mu_{m'} = -\mu_m$; vacuum at the outer face, $\psi_{I+1/2,m} = 0$ for $\mu_m < 0$.

Eigenvalue. Outer iteration l fixes the fission source $Q_{f,i}^{(l)} = \nu\Sigma_{f,i}\phi_i^{(l)}/k^{(l)}$; inner iteration n sweeps $S_i = \Sigma_{s,i}\phi_i^{(n)} + Q_{f,i}^{(l)}$ until ϕ settles; then

$$k^{(l+1)} = k^{(l)} \frac{P^{(l+1)}}{P^{(l)}}, \quad P^{(l)} = \sum_i \nu\Sigma_{f,i} \phi_i^{(l)} V_i \quad (7)$$

Dividing the fission source by k makes a solution exist at *every* system size, so criticality becomes a root search: `brentq` on $k(\text{size}) - 1$, which costs 8 evaluations of k .

The meaning of c . Questions 3 and 4 specify only $c = (\Sigma_s + \nu\Sigma_f)/\Sigma_t$. At $k = 1$ the source bracket collapses to $c\Sigma_t\phi$ whatever the split between scattering and fission, so the critical size depends on c alone and the code takes the simplest split, $\Sigma_t = 1$, $\Sigma_s = 0$, $\nu\Sigma_f = c$. Question 5 has no such freedom and uses the tabulated cross sections directly.

1 Critical radius of a reflected sphere

Question 1 is the one *diffusion* question of this assignment — the S_N machinery above belongs to Questions 3–5 — and the only one with two materials. A multiplying core of radius R_C is wrapped in $d \in \{1, 2, 3, 10\}$ mean free paths of reflector, and the cross sections are again the Question 5 table of Assignment 1.

The two regions

In both regions the substitution $u(r) = r\phi(r)$, which uses $\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\phi}{dr} \right) = \frac{1}{r} \frac{d^2u}{dr^2}$, turns the spherical diffusion equation into a plane one:

$$\frac{d^2u}{dr^2} - \frac{(1-c)\Sigma_t^2}{D_0} u = 0 \quad (8)$$

The core has $c_C > 1$, so the coefficient is negative and the solution oscillates; finiteness at the origin selects the sine. The reflector has $c_R \leq 1$ and decays. With

$$B = k_0(c_C) \Sigma_{t,C}, \quad \kappa = \frac{\Sigma_{t,R}}{\nu_0(c_R)}, \quad R_{\text{ext}} = R_R + \frac{z_0(c_R)}{\Sigma_{t,R}} \quad (9)$$

the two profiles are

$$\phi_C(r) = A \frac{\sin(Br)}{r}, \quad \phi_R(r) = F \frac{\sinh[\kappa(R_{\text{ext}} - r)]}{r} \quad (10)$$

k_0 and ν_0 are Case's discrete eigenvalue on its two branches, the same roots Assignment 1 used: $c \arctan(k_0) = k_0$ above $c = 1$ and $\text{artanh}(1/\nu_0)/(1/\nu_0) = 1/c$ below it.

Matching, and the one criticality equation

The interface at $r = R_C$ carries a flux relation $\phi_R = g \phi_C$ and continuity of the net current $J = -D d\phi/dr$. Since $\phi = u/r$ gives $\phi'/\phi = u'/u - 1/r$, and the two logarithmic derivatives are $u'_C/u_C = B \cot(BR_C)$ and $u'_R/u_R = -\kappa \coth(\kappa L)$ with $L = R_{\text{ext}} - R_C$, all three approximations reduce to a single transcendental equation for R_C :

$$\boxed{D_C B \cot(BR_C) + g D_R \kappa \coth(\kappa L) - \frac{D_C - g D_R}{R_C} = 0} \quad (11)$$

with $\kappa L = [d + z_0(c_R)]/\nu_0(c_R)$, so the reflector enters only through its *optical* thickness. The last term has no slab counterpart: ϕ falls as $1/r$ even where u is flat, so a jump in D at a curved interface drives a current by itself. Equation (11) runs from $+\infty$ at $R_C \rightarrow 0$ to $-\infty$ at $R_C = \pi/B$, the unreflected limit, so the fundamental root is always bracketed by $(0, \pi/B)$ and `brentq` finds it in one call.

The three approximations differ only in what they put into D_0 , z_0 and g :

	(a) continuous classic	(b) continuous asymptotic	(c) discontinuous asymptotic
$D_0(c)$	1/3	$ 1 - c \nu_0^2$	$ 1 - c \nu_0^2$
relaxation rate	$\sqrt{3 1 - c }$	$k_0(c)$ or $1/\nu_0(c)$	$k_0(c)$ or $1/\nu_0(c)$
$z_0(c)$	2/3	$0.710446 [1 + q(1 - c)^2]/c$	$0.710446 [1 + q(1 - c)^2]/c$
$g = \phi_R/\phi_C$	1	1	$\mu_0(c_C)/\mu_0(c_R)$

Table 1: Only the last row makes (c) different from (b). Since $\mu_0 = 1/2$ for every medium in the classic theory, (a) is automatically continuous, and with no reflector at all (b) and (c) are the same bare sphere.

Zimmerman's μ_0 . The discontinuity comes from Zimmerman (UCRL-82792), who writes the partial currents of the asymptotic mode as $\phi_{\pm} = \frac{1}{2}\mu_0\phi_0 \pm \frac{1}{2}\phi_1$, so that $\mu_0\phi_0 = J_+ + J_-$ is the total traffic across a surface and

$$\mu_0(c) = \frac{c\nu_0^2}{2} \ln\left(\frac{\nu_0^2}{\nu_0^2 - 1}\right) \xrightarrow{c>1} \frac{c}{2k_0^2} \ln(1 + k_0^2) \quad (12)$$

the second form being the exact analytic continuation through $\nu_0 = i/k_0$. It is not an ansatz: integrating the discrete mode $\psi(\mu) \propto (\nu_0 - \mu)^{-1}$ over the two half-ranges reproduces it identically, and it returns the two textbook limits $\mu_0 = 1/2$ for an isotropic flux and $\mu_0 = 1$ for a one-way beam. Matching the traffic rather than the flux, $\mu_0(c_C)\phi_C = \mu_0(c_R)\phi_R$, is what makes ϕ itself jump. Nothing is lost at the interface — the same current crosses both faces.

material	c	mfp [cm]	k_0 or $1/\nu_0$	D_0	D [cm]	μ_0	z_0
Pu-239	1.5000	3.064	1.45110	0.23745	0.7275	0.40363	0.47127
U-235	1.3000	3.064	1.05708	0.26847	0.8225	0.43639	0.54552
H ₂ O	0.9000	3.064	0.52543	0.36222	1.1097	0.52660	0.78923
Fe	0.9980	4.300	0.07740	0.33387	1.4356	0.50050	0.71187
Na	1.0000	11.578	0.00000	0.33333	3.8595	0.50000	0.71045

Table 2: Asymptotic parameters of the five materials. Sodium is a pure scatterer, $\Sigma_c = \Sigma_f = 0$, so $c_R = 1$ exactly: ν_0 diverges, $\kappa = 0$, and the reflector solution degenerates to u_R linear in r with $\kappa \coth(\kappa L) \rightarrow 1/L$. The code takes that limit explicitly rather than dividing out a near-zero κ .

Results

The bare spheres of Assignment 1 Question 5, which every reflected radius below is measured against:

core	classic R_c [cm]	asympt. R_c [cm]	classic M_c [kg]	asympt. M_c [kg]
Pu-239	5.8163	5.1890	12.940	9.188
U-235	8.1031	7.4339	42.345	32.696

Table 3: Bare reference. Theory (a) is compared with the classical column, (b) and (c) with the asymptotic one.

reflector	d [mfp]	$R_c^{(a)}$	$R_c^{(b)}$	$R_c^{(c)}$	$M_c^{(a)}$	$M_c^{(b)}$	$M_c^{(c)}$
Water	1	5.3159	4.9484	4.5400	9.879	7.968	6.154
Water	2	5.0855	4.7987	4.3823	8.649	7.267	5.535
Water	3	5.0153	4.7499	4.3321	8.296	7.048	5.347
Water	10	4.9814	4.7244	4.3061	8.129	6.935	5.251
Iron	1	5.4212	4.9575	4.6450	10.478	8.013	6.591
Iron	2	5.1056	4.7364	4.4182	8.753	6.988	5.672
Iron	3	4.9506	4.6237	4.3059	7.979	6.501	5.250
Iron	10	4.6790	4.4188	4.1070	6.737	5.674	4.556
Sodium	1	6.4397	5.7159	5.5171	17.562	12.281	11.044
Sodium	2	6.3134	5.6405	5.4318	16.549	11.801	10.540
Sodium	3	6.2505	5.6022	5.3892	16.059	11.563	10.293
Sodium	10	6.1315	5.5286	5.3080	15.159	11.113	9.835

Table 4: Pu-239 core: critical radius in cm and critical mass in kg, under (a) continuous classic, (b) continuous asymptotic and (c) discontinuous asymptotic diffusion. Bare: 5.8163 cm / 12.940 kg classic, 5.1890 cm / 9.188 kg asymptotic.

reflector	d [mfp]	$R_c^{(a)}$	$R_c^{(b)}$	$R_c^{(c)}$	$M_c^{(a)}$	$M_c^{(b)}$	$M_c^{(c)}$
Water	1	7.2501	6.9113	6.5228	30.330	26.273	22.087
Water	2	6.9185	6.6731	6.2694	26.355	23.649	19.611
Water	3	6.8146	6.5937	6.1865	25.186	22.816	18.844
Water	10	6.7638	6.5520	6.1432	24.627	22.385	18.451
Iron	1	7.2811	6.8392	6.5656	30.720	25.460	22.525
Iron	2	6.8002	6.4625	6.1789	25.027	21.480	18.775
Iron	3	6.5545	6.2644	5.9804	22.411	19.565	17.023
Iron	10	6.1125	5.8970	5.6199	18.176	16.320	14.126
Sodium	1	8.4264	7.7951	7.6194	47.618	37.697	35.205
Sodium	2	8.2322	7.6564	7.4692	44.401	35.721	33.164
Sodium	3	8.1328	7.5844	7.3919	42.811	34.722	32.145
Sodium	10	7.9401	7.4425	7.2414	39.840	32.809	30.221

Table 5: U-235 core, same columns. Bare: 8.1031 cm / 42.345 kg classic, 7.4339 cm / 32.696 kg asymptotic.

Question 1: critical core radius of the reflected spheres

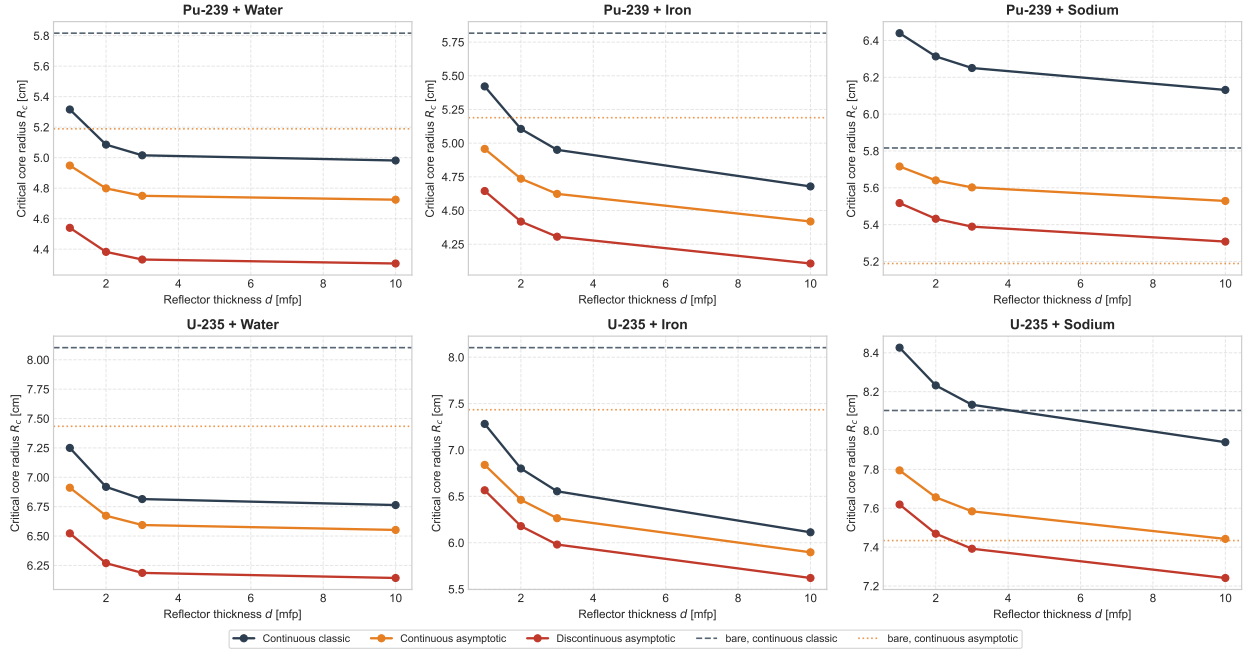


Figure 1: Critical core radius against reflector optical thickness. Dashed and dotted lines are the two bare references. Water saturates by $d \approx 3$ mfp; iron is still improving at $d = 10$; sodium sits above its bare line throughout.

Question 1: critical flux profiles behind 3 mfp of reflector

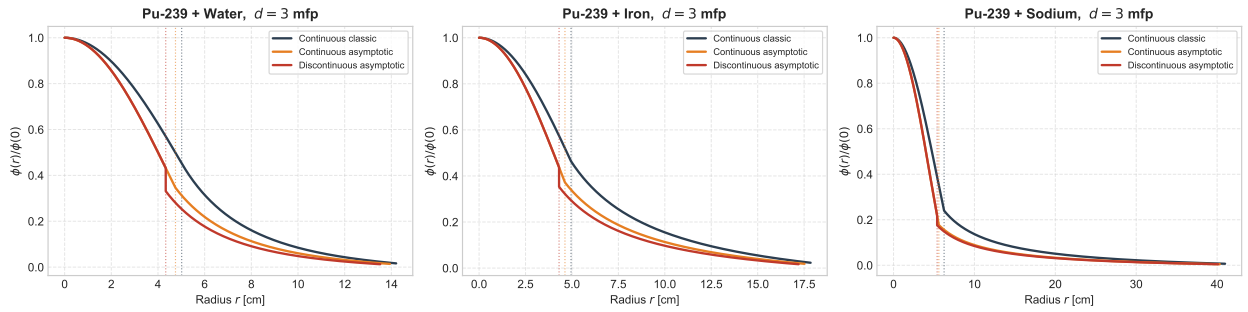


Figure 2: Critical flux across core and reflector for the Pu-239 core behind 3 mfp, normalised to $\phi(0) = 1$; dotted verticals mark each theory's interface. The discontinuous curve drops by the factor g at the interface and is continuous in current, not in flux. Note the horizontal scale of the sodium panel: 3 mfp of sodium is 34.7 cm.

What the three approximations say

The reflector saving. Water and iron behave as reflectors should. Both cores lose 12–17% of their radius behind water and 12–24% behind iron by theory (c), which because the mass goes as R^3 is a factor of nearly two in critical mass: Pu-239 falls from 9.188 kg bare to 4.556 kg behind 10 mfp of iron. The two reflectors saturate differently, and the reason is c_R . Water absorbs ($c_R = 0.90$, $\nu_0 = 1.90$), so a neutron more than about two mean free paths deep never returns and the water columns are flat past $d = 3$. Iron barely absorbs ($c_R = 0.998$, $\nu_0 = 12.9$), so its return keeps improving out to $d = 10$ and beyond — iron starts as the worse reflector at $d = 1$ and ends the better one at $d = 10$.

The ordering of the three theories. At every entry $R_c^{(a)} > R_c^{(b)} > R_c^{(c)}$. The first gap is inherited from the bare problem: the asymptotic $D_0(c_C)$ and $k_0(c_C)$ already give a smaller sphere than $D_0 = 1/3$ does, by 11% for Pu-239. The second is the jump itself. Every pair here has $\mu_0(c_C) < \mu_0(c_R)$, so $g < 1$ — 0.766 for Pu-239 behind water, 0.873 for U-235 behind iron — because the multiplying core has the shorter relaxation length and hence the more forward-peaked angular flux, and so carries more current per unit density. A g below one multiplies down both the leakage term and the curvature term of (11), which is why (c) always predicts the smallest core, by a further 8–9% for water and iron.

The sodium columns are diffusion theory failing. Every sodium radius above is *larger* than the corresponding bare sphere, which no reflector can do. The condition is being solved correctly — (11) was checked against an independent two-region spherical finite-volume k -eigenvalue solve, 3000 cells with harmonic-mean face diffusion coefficients across the material jump, and the two agree to better than 0.4% on exactly these cases. The fault is in the model. Splitting the interface leakage of the Pu-239 core at $d = 10$ mfp into its two parts, $J/\phi = D_R/R_c + D_R/L$:

reflector	D_R [cm]	shell [cm]	geometric D_R/R_c	far-face D_R/L	total vs. bare 0.5592
Water	1.021	30.6	0.2050	0.1826	0.3876
Iron	1.433	43.0	0.3063	0.0381	0.3444
Sodium	3.859	115.8	0.6295	0.0312	0.6607

Table 6: Leakage ratio at the interface, in cm^{-1} , classic theory. Sodium’s far-face return is the best of the three — 10 mfp of a pure scatterer really does send almost everything back — but its geometric term alone exceeds what a bare vacuum surface leaks.

The geometric term is the current driven by the $1/r$ spreading of $\phi = u/r$, and it is harmless only while $D_R \ll R_c$. Sodium’s $\Sigma_t = 0.086368 \text{ cm}^{-1}$ is a mean free path of 11.58 cm against a core radius near 6 cm: a neutron entering the reflector travels two core radii before its first collision, which is precisely the regime Fick’s law expands away from. None of the three approximations repairs it, because a pure scatterer has $D_0 = 1/3$ in all of them and $D_R = 3.86 \text{ cm}$ throughout. Measured against its own bare sphere at $d = 10$, (a) is +5.4%, (b) +6.6% and (c) +2.3% — only the Zimmerman jump helps, and only because $g = 0.807$ multiplies the offending term. Transport theory would not make this mistake: the shell is optically thick and purely scattering, so its true albedo is high and the true critical radius must lie below the bare 5.1890 cm.

3 Critical slab half-thickness by S_N

Reflective at $x = 0$, vacuum at $x = a/2$, 200 cells. The reference column is Assignment 1's exact-transport relation $a/2 = \frac{\pi}{2} \nu_0 - z_0$, evaluated from Case's tabulated ν_0 and z_0 .

c	order	$a/2$ [mfp]	exact transport [mfp]	departure
1.2	S_2	1.48499	1.28981	+15.13%
1.2	S_4	1.31855	1.28981	+2.23%
1.2	S_6	1.29861	1.28981	+0.68%
1.2	S_{10}	1.29218	1.28981	+0.18%
1.5	S_2	0.78001	0.60762	+28.37%
1.5	S_4	0.64949	0.60762	+6.89%
1.5	S_6	0.62089	0.60762	+2.18%
1.5	S_{10}	0.60904	0.60762	+0.23%
1.8	S_2	0.54291	0.39314	+38.09%
1.8	S_4	0.43875	0.39314	+11.60%
1.8	S_6	0.41014	0.39314	+4.32%
1.8	S_{10}	0.39459	0.39314	+0.37%

Table 7: Critical half-thickness in mean free paths. The departure is entirely angular: the same radii are reproduced to seven digits on meshes from 50 to 800 cells.

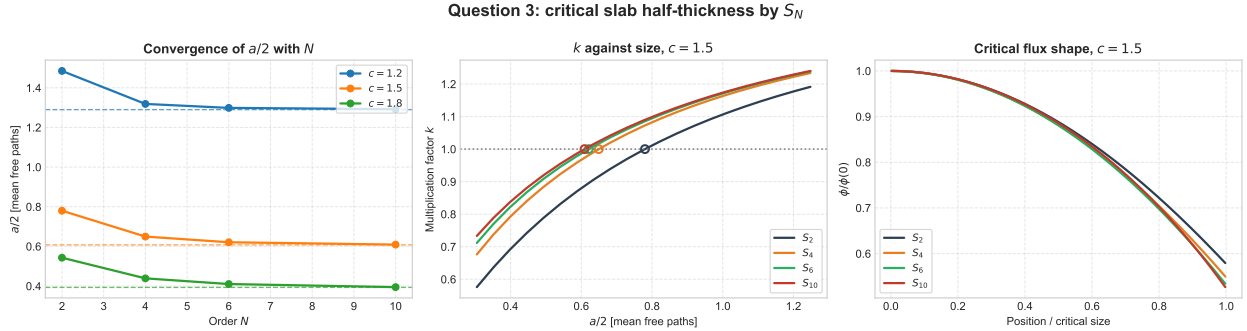


Figure 3: Left: convergence of $a/2$ with N , dashed lines the exact-transport values. Centre: k against half-thickness at $c = 1.5$, circles the $k = 1$ crossings. Right: the normalised critical flux, which barely moves with N — the order affects the boundary, not the interior.

The exact one-speed value at $c = 1.5$ is 0.605055 mfp, the Sood *et al.* Pu-239 slab (1.853722 cm at $\Sigma_t = 0.32640 \text{ cm}^{-1}$, the same table Assignment 1 Question 5 uses). The S_N sequence walks down onto it monotonically: 0.609042 at S_{10} , 0.606407 at S_{16} , 0.605631 at S_{24} , 0.605195 at S_{48} .

P_N against S_{N+1}

In slab geometry the two are algebraically the same interior equations. Expanding ψ in $N + 1$ Legendre moments and truncating gives the same $N + 1$ coupled ODEs as collocating at the $N + 1$ Gauss–Legendre points, because a Gauss–Legendre rule of order $N + 1$ integrates polynomials up to degree $2N + 1$ exactly. **Any difference between P_N and S_{N+1} is therefore a difference of boundary condition**, and it is largest at low N where the boundary is the whole story:

- S_{N+1} sets $\psi(a/2, \mu_m) = 0$ on each incoming ordinate — the *Mark* condition.
- P_N needs conditions on moments instead. *Marshak*'s, $\int_0^1 P_{2j+1}(\mu)\psi d\mu = 0$, weight the whole incoming half-range; Mark's pin the same isolated directions S_{N+1} does.

With Mark conditions the two agree; with Marshak they do not. Concretely at $c = 1.5$: S_2 above gives $a/2 = 0.78001$, and P_1 with the Mark extrapolation $l_0 = 1/\sqrt{3}$ gives $\arctan(1/Bl_0)/B = 0.78001$ — the same five digits — while P_1 with the Marshak $l_0 = 2/3$ gives 0.72348, a 7% spread between two methods that share every interior equation.

A second source of difference is that S_N is a collocation method: it satisfies the equation exactly along N directions and says nothing between them. At the vacuum face the leaked current $\sum_{\mu_m > 0} w_m \mu_m \psi_m$ is that of a flux strongly peaked towards $\mu = 1$, which a coarse quadrature underestimates — so the slab has to be made *thicker* than it should be to stay critical, and the table comes down from above.

All of that is a statement about *slabs*. In the sphere the equivalence fails at the interior equations themselves: the angular derivative $\frac{1-\mu^2}{r} \frac{\partial \psi}{\partial \mu}$ couples the Legendre moments in a different pattern from the way the α recursion of (5) couples the ordinates, so P_1 and S_2 are genuinely different approximations there rather than the same one under two boundary conditions. At $c = 1.5$, S_2 gives $R_c = 1.607$ mean free paths where P_1 with the Mark extrapolation gives 1.919 and with Marshak's 1.822 — a spread no choice of boundary condition can close.

4 Critical sphere radius by S_N

Reflective at $r = 0$, vacuum at $r = R$, 100 cells, with the angular redistribution term of (5). The reference is Assignment 1's $\Sigma_t R_c = \pi \nu_0 - z_0$.

c	order	R_c [mfp]	exact transport [mfp]	departure
1.2	S_2	3.06060	3.17204	−3.51%
1.2	S_4	3.14631	3.17204	−0.81%
1.2	S_6	3.15991	3.17204	−0.38%
1.2	S_{10}	3.16728	3.17204	−0.15%
1.5	S_2	1.60725	1.69010	−4.90%
1.5	S_4	1.66739	1.69010	−1.34%
1.5	S_6	1.67934	1.69010	−0.64%
1.5	S_{10}	1.68595	1.69010	−0.25%
1.8	S_2	1.11753	1.18295	−5.53%
1.8	S_4	1.16413	1.18295	−1.59%
1.8	S_6	1.17412	1.18295	−0.75%
1.8	S_{10}	1.17971	1.18295	−0.27%

Table 8: Critical radius in mean free paths. Note the sign: the sphere converges from *below*, the slab from above.

Cross-check between the two geometries. Both drive the asymptotic flux to zero at the same extrapolated surface, so $a/2 + z_0 = \frac{\pi}{2} \nu_0$ and $R_c + z_0 = \pi \nu_0$, and therefore

$$z_0 = R_c - 2(a/2) \tag{13}$$

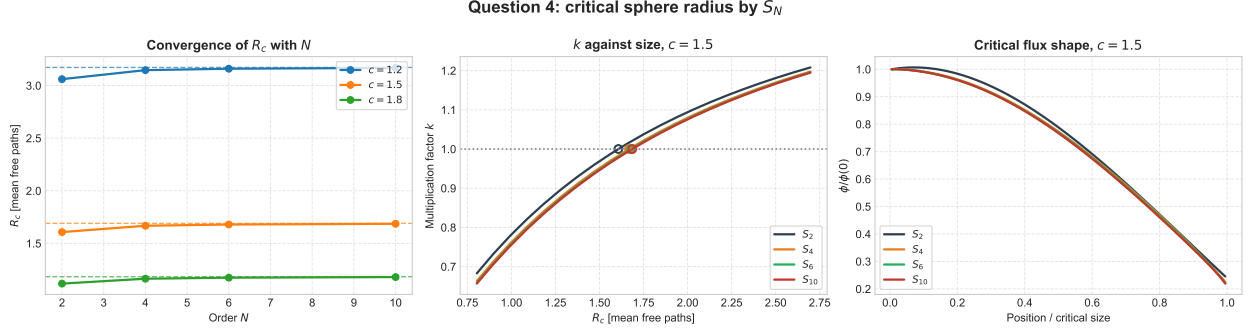


Figure 4: The same three panels for the sphere. The S_2 curve is the outlier in all three: in curvilinear geometry S_2 is not the P_1 equation, and the angular redistribution term it misrepresents is a loss from the interior directions.

which lets the two tables above check each other without any further calculation:

c	from S_2	from S_4	from S_6	from S_{10}	Case's table
1.2	0.091	0.509	0.563	0.583	0.592
1.5	0.047	0.368	0.438	0.468	0.475
1.8	0.032	0.287	0.354	0.391	0.397

Table 9: Extrapolation distance z_0 implied by the two geometries, in mean free paths. At S_{10} it is 1.5% below Case's value at all three c — the same shortfall three times, which is the relation's own asymptotic error. At S_2 it nearly vanishes, which is the sharpest statement of what low-order S_N gets wrong: the boundary, not the interior.

5 Critical mass of U-235 and Pu-239 by S_N

The cross sections are the Question 5 table of Assignment 1 — $\Sigma_t = 0.32640 \text{ cm}^{-1}$ for both, giving $c = 1.500$ for Pu-239 and $c = 1.300$ for U-235 — and the critical mass follows from the converged radius as $M_c = \frac{4}{3}\pi R_c^3 \rho$. Here $\Sigma_s > 0$, so the inner iteration does real work: about ten sweeps per outer, against one for the c -only problems above. The orders are S_2 to S_{16} rather than the S_2 – S_{10} of the previous questions, since the mass goes as the cube of the radius and is the more demanding quantity.

Diffusion overestimates both radii, and by more for Pu-239 than for U-235: the surface sits about two mean free paths from the centre in both cases, but the higher c makes the Pu-239 flux fall off faster and its surface angular flux more forward-peaked, which is exactly what a diffusion approximation cannot represent. S_{16} recovers the difference to within 0.3% of the transport-theory radius.

material	order	c	R_c [cm]	M_c [kg]	asympt. diffusion R_c [cm]	departure
Pu-239	S_2	1.500	4.9242	7.852	5.1890	-5.10%
Pu-239	S_4	1.500	5.1084	8.767	5.1890	-1.55%
Pu-239	S_8	1.500	5.1586	9.028	5.1890	-0.59%
Pu-239	S_{16}	1.500	5.1730	9.104	5.1890	-0.31%
U-235	S_2	1.300	7.1218	28.748	7.4339	-4.20%
U-235	S_4	1.300	7.3514	31.619	7.4339	-1.11%
U-235	S_8	1.300	7.4072	32.345	7.4339	-0.36%
U-235	S_{16}	1.300	7.4231	32.553	7.4339	-0.15%

Table 10: Critical radius and mass, against the asymptotic diffusion result of Assignment 1 Question 5. The mass is the cube of the radius, so the 5% radius error of S_2 is a 14% error in mass.

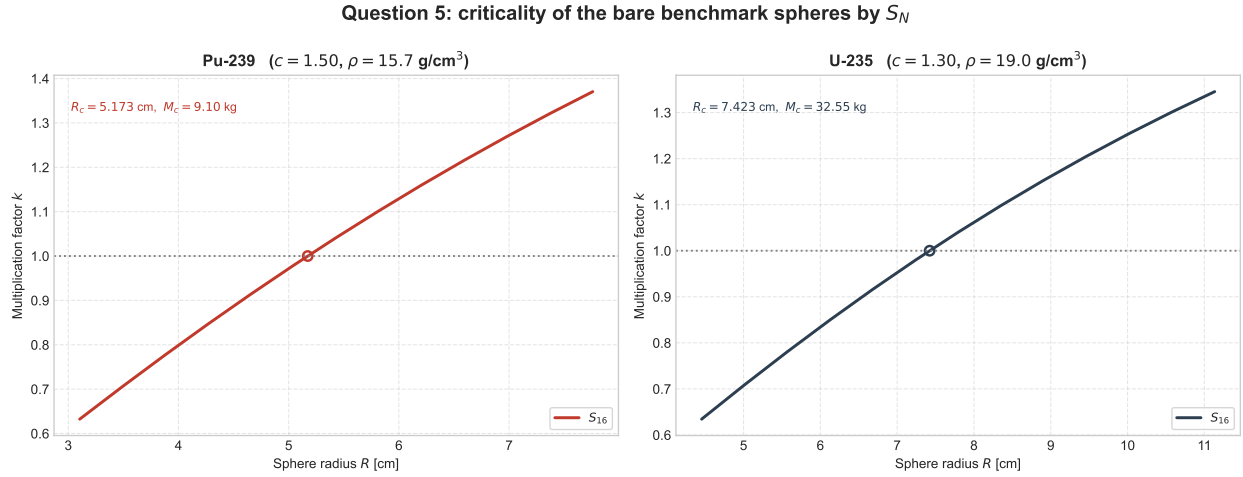


Figure 5: k against sphere radius at S_{16} , with the $k = 1$ crossing that fixes the critical mass.